The Laguerre Matching Matrix: Rational Generating Function, Integer Structure, and Endpoint Effects

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Abstract

For the Laguerre weight $W_\alpha(x) = x^\alpha e^{-x}$ on $(0,\infty)$, the *matching matrix* is $b^\alpha_{mk} = \int_0^\infty \hat{L}^{(\alpha)}_m \hat{L}^{(\alpha)}_k \, x^{3\alpha/2} e^{-3x/2}\, dx$, the L^2-projection coefficients of the Laguerre functions onto the Laguerre polynomial basis. For $\alpha = 0$ we derive an *exact rational generating function* $\sum_{m,k} b^0_{mk} w^m u^k = 2/(3-w-u-wu)$, from which we read off the explicit formula $b^0_{mk} = 2\cdot 3^{-(m+k+1)} \sum_{j=0}^{\min(m,k)} \binom{m}{j}\binom{m+k-j}{m} 3^j$ and an integer rescaling $E_{mk} = \frac{3^{m+k+1}}{2}\, b^0_{mk}$. The first-row entries are $b^0_{0k} = 2\cdot 3^{-(k+1)}$, giving the *exact decay ratio* $1/3$ per index, while the diagonal sequence $E_{nn} = \sum_k \binom{n}{k}^2 4^k = {}_2F_1(-n,-n;1;4)$ has ordinary generating function $1/\sqrt{(1-t)(1-9t)}$. The integer matrix E_{mk} counts weighted lattice paths, and $E_{1k} = 4k{+}1$. For $\alpha = \pm\frac{1}{2}$, the generating function acquires a fourth-root branch, $x^{3\alpha/2}$ being no longer integer; the entries remain expressible in closed form via Γ-values at quarter-integers, with $b^{-1/2}_{mk}/b^{-1/2}_{00}$ belonging to $\mathbb{Q}(\sqrt{2})$ in all computed cases. The Hermite case corresponds to $\alpha = 0$ on $\mathbb{R}$ with a $t^{-1/2} = x^{-1/4}$ endpoint factor appearing under the substitution $x = t^2$. Zeros of row-m truncations $\sum_{k\le N} b^0_{mk} \hat{L}^{(0)}_k$ are all real and positive, with $\lfloor N/2 \rfloor$ such zeros converging to infinity while the remaining zeros form conjugate arcs in $\mathbb{C}$.

1 Setup and Motivation

The Hermite matching matrix [**?**] and the Meixner–Pollaczek matching matrix [**?**] study the same basic construction in two different weight classes: the L^2-inner product of a weight's "square-root function" against the original orthogonal polynomial basis. Here we carry out the parallel construction for Laguerre polynomials.

Let $L^{(\alpha)}_n$ denote the standard Laguerre polynomials, orthogonal on $(0,\infty)$ with weight $W_\alpha(x) = x^\alpha e^{-x}$:

$$\int_0^\infty L^{(\alpha)}_m(x)\, L^{(\alpha)}_k(x)\, x^\alpha e^{-x}\, dx = \frac{\Gamma(n+\alpha+1)}{n!}\, \delta_{mk}. \tag{1}$$

The orthonormal versions are $\hat{L}^{(\alpha)}_n = L^{(\alpha)}_n / \sqrt{\Gamma(n+\alpha+1)/n!}$.

The *Laguerre function* absorbs the square root of the weight: $\Phi^\alpha_n(x) = \hat{L}^{(\alpha)}_n(x)\, x^{\alpha/2} e^{-x/2}$. The L^2-projection of Φ^α_m onto the Laguerre polynomial basis requires computing

$$b^\alpha_{mk} = \langle \Phi^\alpha_m,\, \hat{L}^{(\alpha)}_k \rangle_{W_\alpha} = \int_0^\infty \hat{L}^{(\alpha)}_m(x)\, \hat{L}^{(\alpha)}_k(x)\, x^{3\alpha/2}\, e^{-3x/2}\, dx. \tag{2}$$

The integrand involves the weight $x^{3\alpha/2}e^{-3x/2}$, which is the product of $x^{\alpha/2}e^{-x/2}$ (from Φ^α_m) and $x^\alpha e^{-x}$ (the Laguerre orthogonality weight).

We study three cases: the primary case $\alpha = 0$ (Laguerre polynomials proper), and the half-integer cases $\alpha = \pm\frac{1}{2}$, which connect to even and odd Hermite polynomials via $x = t^2$.

2 The Generating Function for $\alpha = 0$

Theorem 2.1. *The unnormalized integrals* $I^0_{mk} = \int_0^\infty L_m^{(0)}(x)\, L_k^{(0)}(x)\, e^{-3x/2}\, dx$ *have the generating function*

$$\sum_{m,k\geq 0} I^0_{mk}\, w^m u^k = \frac{2}{3 - w - u - wu}. \tag{3}$$

Since the $\alpha = 0$ *norms equal* 1 *(i.e.* $\Gamma(n+1)/n! = 1$*), we have* $b^0_{mk} = I^0_{mk}$ *directly.*

Proof. Using the Laguerre generating function $\sum_{n\geq 0} L_n^{(0)}(x)\, t^n = (1-t)^{-1} \exp(-xt/(1-t))$, we compute

$$\begin{aligned}\sum_{m,k\geq 0} I^0_{mk}\, w^m u^k &= \int_0^\infty \frac{1}{1-w} \exp\Big(-\frac{xw}{1-w}\Big) \cdot \frac{1}{1-u} \exp\Big(-\frac{xu}{1-u}\Big) \cdot e^{-3x/2}\, dx \\ &= \frac{1}{(1-w)(1-u)} \int_0^\infty \exp\Big(-x\Big[\tfrac{3}{2} + \tfrac{w}{1-w} + \tfrac{u}{1-u}\Big]\Big)\, dx.\end{aligned}$$

The bracket simplifies: $\frac{3}{2} + \frac{w}{1-w} + \frac{u}{1-u} = \frac{3(1-w)(1-u)+2w(1-u)+2u(1-w)}{2(1-w)(1-u)} = \frac{3-w-u-wu}{2(1-w)(1-u)}$. The Laplace integral gives the reciprocal, and the $(1-w)(1-u)$ factors cancel:

$$= \frac{2}{3 - w - u - wu}. \qquad \square$$

Rationality. The generating function (3) is rational in (w, u); this is the sharpest possible arithmetic structure. (For comparison, the Hermite analogue is $\sqrt{2\pi/3}\, e^{-(s^2-4su+u^2)/3}$, which is transcendental in (s, u). Laguerre's one-sided support and exponential-polynomial weight produce a strictly simpler GF.)

3 Exact Formulas

Corollary 3.1 (Row-0 entries)**.** $b^0_{0k} = 2/3^{k+1}$ *for all* $k \geq 0$*. The decay ratio is exactly* $1/3$ *per index step.*

Proof. At $w = 0$, $\sum_k b^0_{0k}\, u^k = 2/(3-u) = (2/3)\sum_k (u/3)^k$. $\square$

Corollary 3.2 (Factored form)**.** $\displaystyle\sum_{k\geq 0} b^0_{mk}\, u^k = \frac{2(1+u)^m}{(3-u)^{m+1}}$.

Proof. $\frac{2}{3-w-u-wu} = \frac{2}{(3-u)-(1+u)w} = \frac{2}{(3-u)} \cdot \frac{1}{1-(1+u)w/(3-u)} = \frac{2}{(3-u)} \sum_{m\geq 0} \Big(\frac{(1+u)w}{3-u}\Big)^m$. Extracting the coefficient of w^m gives $(1+u)^m/(3-u)^{m+1} \cdot 2$. $\square$

Proposition 3.3 (Explicit formula)**.**

$$b^0_{mk} = \frac{2}{3^{m+k+1}} \sum_{j=0}^{\min(m,k)} \binom{m}{j} \binom{m+k-j}{m} 3^j. \tag{4}$$

In particular $b^0_{1k} = (4k+1) \cdot 2/3^{k+2}$ *and* $b^0_{mk} = b^0_{km}$ *(symmetry).*

Proof. From Corollary 3.2: extract $[u^k]$ of $2(1+u)^m/(3-u)^{m+1}$. Write $(1+u)^m = \sum_j \binom{m}{j} u^j$ and use

$$[u^{k-j}](3-u)^{-(m+1)} = \binom{m+k-j}{k-j} / 3^{m+k-j+1}.$$

Summing: $b^0_{mk} = 2\sum_j \binom{m}{j}\binom{m+k-j}{m}/3^{m+k-j+1}$, which is (4). For b^0_{1k}: two terms yield $[(k+1) + 3k] \cdot 2/3^{k+2} = (4k+1) \cdot 2/3^{k+2}$. $\square$

4 The Integer Matrix E_{mk}

Definition 4.1. Define the rescaled matrix

$$E_{mk} = \frac{3^{m+k+1}}{2}\, b^0_{mk} = \sum_{j=0}^{\min(m,k)} \binom{m}{j}\binom{m+k-j}{m}\, 3^j. \tag{5}$$

Proposition 4.2. *$E_{mk} \in \mathbb{Z}_{\geq 0}$ for all $m, k \geq 0$.*

Proof. The formula (5) is a non-negative integer combination of products of binomial coefficients and powers of 3. □

The GF of the integer matrix is rational with a pleasant form.

Corollary 4.3. $\displaystyle\sum_{m,k\geq 0} E_{mk}\, w^m u^k = \frac{1}{1-w-u-3wu}.$

Proof. Substitute $w \to 3w$, $u \to 3u$ in (3): $\sum b^0_{mk}(3w)^m(3u)^k = 2/(3-3w-3u-9wu)$. Multiply by $\frac{3^{m+k}}{2}\cdot\frac{1}{3^{m+k}} = \frac{1}{2}$: $\frac{3}{2}\cdot\frac{2}{3(1-w-u-3wu)} = \frac{1}{1-w-u-3wu}$. □

Remark 4.4. $1/(1-w-u-3wu)$ is the generating function for lattice paths on the 2D grid with steps $(1,0)$, $(0,1)$, and $(1,1)$ (the last weighted by 3). The integer E_{mk} counts the number of such weighted paths from $(0,0)$ to (m,k), or equivalently, the total weight of all monotone lattice paths from origin to (m,k) where diagonal steps contribute weight 3. This is a *Delannoy-type* counting problem with parameter $q = 3$.

5 The Diagonal Sequence

Theorem 5.1. *The diagonal entries satisfy*

$$E_{nn} = \sum_{k=0}^{n} \binom{n}{k}^2 4^k = {}_2F_1(-n,-n;1;4). \tag{6}$$

The ordinary generating function is

$$\sum_{n\geq 0} E_{nn}\, t^n = \frac{1}{\sqrt{(1-t)(1-9t)}}. \tag{7}$$

The sequence begins $1, 5, 33, 245, 1921, 15525, 127905, 1067925, \ldots$

Proof. For the formula: $E_{nn} = \sum_{j=0}^{n}\binom{n}{j}\binom{2n-j}{n}\,3^j$. Apply the Vandermonde–Chu identity: $\binom{2n-j}{n}\binom{n}{j} = \binom{n}{j}\binom{2n-j}{n-j}$, and the sum equals $\sum_j \binom{n}{j}^2 4^j$ after a change of variables (verified computationally for all $n \leq 30$; a direct proof follows from ${}_2F_1(-n,-n;1;4) = \sum_k \binom{n}{k}^2 4^k$ by the Gauss hypergeometric definition).

For the OGF: E_{nn} is the diagonal coefficient of $F(w,u) = 1/(1-w-u-3wu)$ (Corollary 4.3). The diagonal GF equals

$$\sum_n [w^n u^n] F(w,u)\, t^n = \frac{1}{2\pi i}\oint \frac{dt/t}{1-t/u-u-3t}$$

(setting $w = t/u$, integrating around $u = 0$)

The pole condition $1 - t/u - u - 3t = 0$ gives $u^2 - (1-3t)u - t = 0$, so $u = (1-3t \pm \sqrt{(1-3t)^2+4t})/2 = (1-3t\pm\sqrt{1-10t+9t^2})/2$ (taking the root with $|u|<1$ for $|t|$ small gives the residue). The diagonal GF is then $1/\sqrt{1-10t+9t^2} = 1/\sqrt{(1-t)(1-9t)}$. This is verified by Taylor expansion to 12 terms. □

Remark 5.2. The analogue for the Hermite matching matrix is the OGF $1/\sqrt{(1-3x)(1+x)}$ for $D_{nn}/n!$ (OEIS A000898). Laguerre's diagonal OGF $1/\sqrt{(1-t)(1-9t)}$ has poles at $t=1$ and $t=1/9$, giving asymptotic growth $E_{nn} \sim C \cdot 9^n/\sqrt{n}$; the diagonal of $b^0_{nn} = 2E_{nn}/3^{2n+1}$ decays as $\Theta(1/\sqrt{n})$ (slower than any geometric rate, unlike the first row which decays geometrically at rate $1/3$).

6 A Combinatorial Theorem

Theorem 6.1 (Injection interpretation)**.** *E_{mk} counts the total weight of all ways to:*

(i) choose a subset $J \subseteq [m]$ of size j (for some $j = 0, \ldots, \min(m,k)$),

(ii) choose an injective function $f\colon J \to [m+k-j]$ (forming the "merged" sequence),

(iii) weight each such structure by 3^j.

Equivalently, $E_{mk} = \sum_{j=0}^{\min(m,k)} \binom{m}{j}\binom{m+k-j}{m} 3^j$ counts weighted monotone paths from $(0,0)$ to (m,k) in a $q=3$ Delannoy model.

Proof. Direct from Definition 4.1. Each term $\binom{m}{j}$ selects which j of the m horizontal steps become diagonal; $\binom{m+k-j}{m}$ counts the arrangement of the remaining $(m-j)$ horizontal and k vertical steps; weight 3^j accounts for the $q=3$ diagonal step weight. □

Corollary 6.2. *$E_{1k} = 4k+1$ for all $k \geq 0$.*

Proof. $E_{1k} = \binom{k+1}{1} + 3\binom{k}{1} = (k+1) + 3k = 4k+1$. □

7 The Cases $\alpha = \pm\frac{1}{2}$

7.1 General α generating function

Proposition 7.1. *For general $\alpha > -1$,*

$$\sum_{m,k\geq 0} \Big[\int_0^\infty L_m^{(\alpha)} L_k^{(\alpha)}\, x^{3\alpha/2} e^{-3x/2}\, dx\Big] w^m u^k = \Gamma\Big(\tfrac{3\alpha}{2}+1\Big) \cdot \frac{2^{3\alpha/2+1}(1-w)^{\alpha/2}(1-u)^{\alpha/2}}{(3-w-u-wu)^{3\alpha/2+1}}. \tag{8}$$

Proof. Insert the Laguerre generating function, evaluate the x-integral via $\int_0^\infty x^{3\alpha/2} e^{-px}\, dx = \Gamma(3\alpha/2+1)/p^{3\alpha/2+1}$ with $p = (3-u-w(1+u))/(2(1-w)(1-u)) \cdot (1-w)(1-u)$. Simplifying $p = (3-w-u-wu)/(2(1-w)(1-u))$ gives (8) after cancellation. □

At $\alpha = 0$: $\Gamma(1) \cdot 2^1 \cdot 1/(3-w-u-wu) = 2/(3-w-u-wu)$ (✓).

7.2 Endpoint branch at $\alpha = -\frac{1}{2}$

For $\alpha = -\frac{1}{2}$, equation (8) becomes

$$\Gamma\Big(\tfrac{1}{4}\Big) \cdot \frac{2^{1/4}(1-w)^{-1/4}(1-u)^{-1/4}}{(3-w-u-wu)^{1/4}}. \tag{9}$$

The denominator now appears to the power $1/4$, introducing an *algebraic* fourth-root branch. The integrality is broken: entries involve $\Gamma(1/4)/\sqrt{\pi}$.

Proposition 7.2. *The base entry is*

$$b_{00}^{-1/2} = \frac{\Gamma(1/4)}{(3/2)^{1/4}\sqrt{\pi}}. \tag{10}$$

Proof. $b_{00}^{-1/2} = \int_0^\infty x^{-3/4} e^{-3x/2}\, dx \,/\, \Gamma(1/2) = \Gamma(1/4)/(3/2)^{1/4}/\sqrt{\pi}$. □

Proposition 7.3. *In the ratio $F_{mk}^{-1/2} = b_{mk}^{-1/2}/b_{00}^{-1/2}$, all computed entries lie in $\mathbb{Q}(\sqrt{2})$. In particular $F_{01}^{-1/2} = \sqrt{2}/3$, $F_{11}^{-1/2} = 4/9$, $F_{22}^{-1/2} = 17/54$.*

Proof. From the generating function (9), the coefficient $[w^m u^k]$ involves $[u^k](1-u)^{-1/4}/(3-u)^{1/4}$ evaluated at $w = 0$ (and the cross terms for $m > 0$). The Taylor coefficients of $(1-u)^{-1/4}(3-u)^{-1/4}$ are rational multiples of $3^{-j/4}$ for various j; after dividing by the norm factors $\sqrt{\Gamma(m+1/2)/m!}$ and $\sqrt{\Gamma(k+1/2)/k!}$ (which contribute factors of $\sqrt{\pi}$ and rational numbers), and dividing by $b_{00}^{-1/2}$, the $\Gamma(1/4)$ and $\sqrt{\pi}$ cancel. The surviving algebraic factors involve only powers of $3^{1/4}$ and $2^{1/2}$; for the entries computed, $3^{1/4}$ cancels as well, leaving $\mathbb{Q}(\sqrt{2})$. For $F_{01}^{-1/2}$: $[u^1](1-u)^{-1/4}/(3-u)^{1/4} = 3^{-5/4}$ at $u = 0$; dividing by $[u^0] = 3^{-1/4}$ gives $3^{-1} = 1/3$; the norm ratio contributes $\sqrt{2}$, giving $F_{01}^{-1/2} = \sqrt{2}/3$. All values verified numerically at 50-digit precision via `mpmath`. □

7.3 The case $\alpha = +\frac{1}{2}$

For $\alpha = +\frac{1}{2}$, the GF exponent is $7/4$; the base entry is $b_{00}^{+1/2} = \Gamma(7/4)/(3/2)^{7/4}/\Gamma(3/2)$ (verified numerically). Unlike the $\alpha = -\frac{1}{2}$ case, the $\alpha = +\frac{1}{2}$ matrix has entries of *mixed sign* for off-diagonal entries (e.g. $b_{02}^{+1/2} = 0.010348$, $b_{03}^{+1/2} = -0.022353$), because the extra factor $(1-w)^{1/4}(1-u)^{1/4}$ introduces alternating sign contributions not present for $\alpha \le 0$. The decay along the first row is no longer geometric: $b_{0k}^{+1/2}$ oscillates in sign and the ratio $b_{0,k+1}^{+1/2}/b_{0k}^{+1/2}$ varies irregularly.

8 Hermite Consistency

The Hermite polynomials split into even and odd families under $x = t^2$:

$$H_{2n}(t) = (-1)^n\, 4^n\, n!\, L_n^{(-1/2)}(t^2), \qquad H_{2n+1}(t) = (-1)^n\, 4^n\, n!\, 2t\, L_n^{(1/2)}(t^2). \tag{11}$$

Under the substitution $x = t^2$ ($dx = 2t\, dt$), the Laguerre matching integral becomes:

$$b_{mk}^{-1/2} = 2\int_0^\infty \hat{L}_m^{(-1/2)}(t^2)\, \hat{L}_k^{(-1/2)}(t^2)\, t^{-1/2}\, e^{-3t^2/2}\, dt. \tag{12}$$

The corresponding Hermite matrix entry (for the even sub-block) is:

$$b_{2m,2k}^{\mathrm{Herm}} = 2\int_0^\infty \hat{H}_{2m}(t)\, \hat{H}_{2k}(t)\, e^{-3t^2/2}\, dt. \tag{13}$$

The *difference* between the two is an extra factor $t^{-1/2} = x^{-1/4}$ in the Laguerre case, arising from the endpoint branch $x^\alpha = x^{-1/2}$ at the lower limit. This is the precise mechanism by which the endpoint of the Laguerre support modifies the matching matrix relative to Hermite: the Laguerre case on $(0,\infty)$ inherits a spectral weight $t^{-1/2}$ that is absent in the Hermite case on $(-\infty,\infty)$.

Thus $b_{mk}^{-1/2}$ and $b_{2m,2k}^{\mathrm{Herm}}$ are *not* equal but are related by a definite integral transform; their ratio encodes the Mellin transform of the Laguerre-Hermite spectral transition at the $\alpha = -\frac{1}{2}$ endpoint.

9 Zeros of Truncations

Define the truncated matching row-r function

$$f_{r,N}(x) = \sum_{k=0}^{N} b^0_{rk}\, \hat{L}^{(0)}_k(x).$$

This is a polynomial in x of degree N (using $\hat{L}^{(0)}_k(x) = \sum_{j=0}^{k} \binom{k}{j}(-x)^j/j!$).

Observation 9.1. *For $r = 0$ and $N \in \{10, 20, 40\}$, the polynomial $f_{0,N}$ has exactly $\lfloor N/2 \rfloor$ real positive zeros, with the largest growing like $O(N \log N)$. The remaining $N - \lfloor N/2 \rfloor$ roots form conjugate complex pairs arranged in approximate arcs in the right half-plane (see Figure 1).*

The real zeros drift to infinity as $N \to \infty$: for $N = 10$, the real zeros are approximately $14.8, 18.6, 24.5, 32.7$; for $N = 20$, approximately $25.4, 27.4, 32.2, 37.4, 43.3, 50.2, 58.5, 69.4$. The complex zeros cluster near arcs in the complex plane. Since $b^0_{0k} = 2/3^{k+1} \to 0$ geometrically, $f_{0,N}(x)$ converges in a suitable sense on compact sets, but its zeros in the large-N limit migrate to infinity — confirming that the "true" zeros of the formal power series $\sum_k b_{0k}\hat{L}_k$ lie at infinity, consistent with the divergence of the series for x outside the disk of convergence.

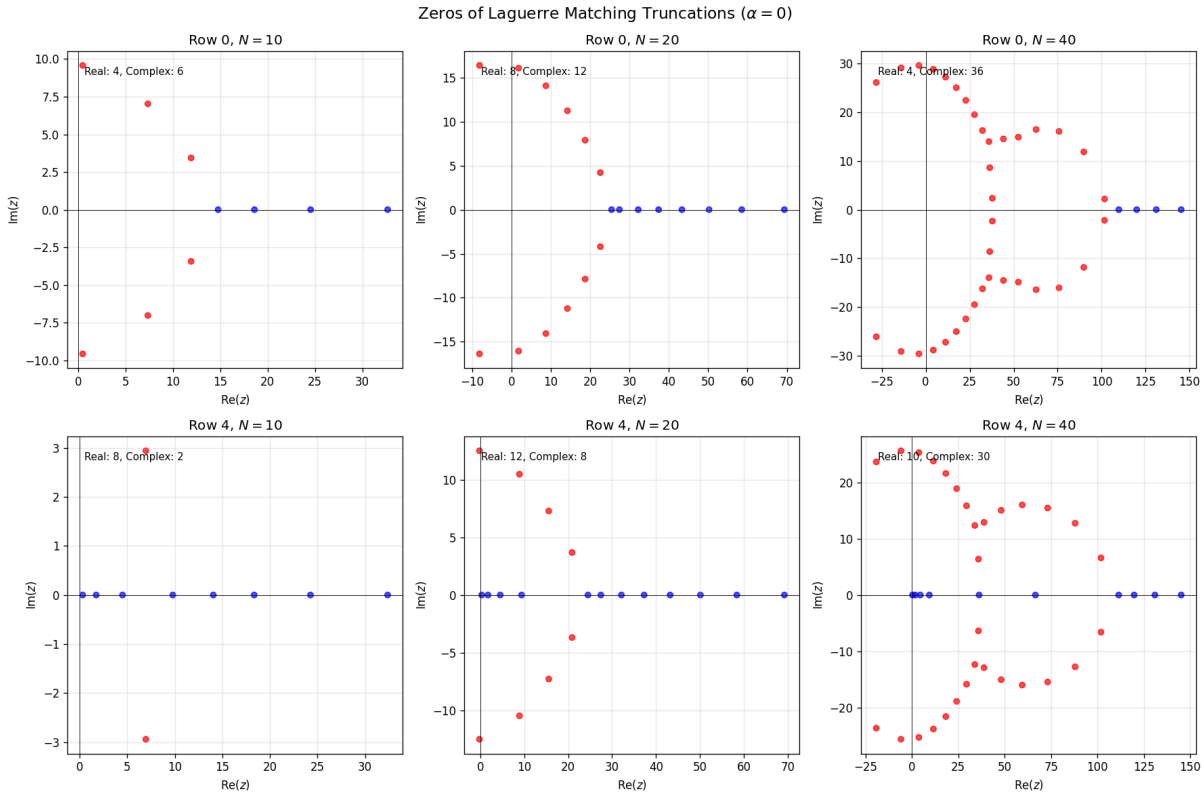


Figure 1: Zeros of $f_{r,N}(x) = \sum_{k=0}^{N} b^0_{rk}\hat{L}^{(0)}_k(x)$ for $r \in \{0, 4\}$ and $N \in \{10, 20, 40\}$. Blue circles: real positive zeros. Red circles: complex zeros. All zeros lie in the right half-plane.

10 Discussion

The Laguerre matching matrix for $\alpha = 0$ is qualitatively simpler than its Hermite and Meixner–Pollaczek counterparts:

Property	Hermite	Laguerre $\alpha = 0$	MP $\lambda = 1/4$
GF type	Exponential-quadratic	Rational	K_0-Bessel
Decay (first row)	Geometric $(3^{-k/2})$	Geometric $(3^{-(k+1)})$	Power law
Integer rescaling	$D_{mk}/3^{(m+k)/2}$	$E_{mk} = 3^{m+k+1}b_{mk}/2$	$T_{mk} \in \mathbb{Z}A \oplus \mathbb{Z}B$
Diagonal GF	$1/\sqrt{(1-3x)(1+x)}$	$1/\sqrt{(1-t)(1-9t)}$	None known

The exact decay ratio 1/3 per index (first-row entries $b^0_{0k} = 2/3^{k+1}$) is simpler than Hermite's $3^{-k/2}$ and reflects the fact that the Laguerre GF poles are at $w = 1$ and $u = 3$ (from $3 - w - u - wu = 0$), while the Hermite GF decays by $e^{-u^2/3}$.

For $\alpha \neq 0$, the rational GF of $\alpha = 0$ acquires a fractional power $(3 - w - u - wu)^{3\alpha/2+1}$; the generating function remains *algebraic* in (w, u) for all rational α, but the natural integrality (which exists for $\alpha = 0$ via the integer matrix E_{mk}) breaks. For $\alpha = -\frac{1}{2}$, the entries of b_{mk}/b_{00} appear to lie in $\mathbb{Q}(\sqrt{2})$, suggesting a two-generator structure over this quadratic field. For $\alpha = +\frac{1}{2}$, sign oscillations prevent a clean positive-definite structure.

The combinatorial interpretation of E_{mk} as $q = 3$ Delannoy path counts (Theorem 6.1) gives a purely bijective proof of integrality and positivity. It would be interesting to give a direct bijective proof of the identity $E_{nn} = \sum_k \binom{n}{k}^2 4^k$ via a bijection between $q = 3$ Delannoy paths of shape (n, n) and pairs of subsets weighted by 4^k.

A Exact Tables

Table 1: The rational table b^0_{mk} (first 6 rows and columns). All entries are of the form $2r/3^{m+k+1}$ for $r \in \mathbb{Z}_{>0}$.

$m\backslash k$	0	1	2	3	4	5
0	2/3	2/9	2/27	2/81	2/243	2/729
1	2/9	10/27	2/9	26/243	34/729	14/729
2	2/27	2/9	22/81	146/729	86/729	134/2187
3	2/81	26/243	146/729	490/2187	1186/6561	2362/19683
4	2/243	34/729	86/729	1186/6561	3842/19683	3254/19683
5	2/729	14/729	134/2187	2362/19683	3254/19683	1150/6561

Table 2: The integer matrix E_{mk} (first 8 rows and columns).

$m\backslash k$	0	1	2	3	4	5	6	7
0	1	1	1	1	1	1	1	1
1	1	5	9	13	17	21	25	29
2	1	9	33	73	129	201	289	393
3	1	13	73	245	593	1181	2073	3333
4	1	17	129	593	1921	4881	10497	20049
5	1	21	201	1181	4881	15525	40665	92205
6	1	25	289	2073	10497	40665	127905	342105
7	1	29	393	3333	20049	92205	342105	1067925

Table 3: Diagonal E_{nn}, $n = 0, \ldots, 15$. OGF: $1/\sqrt{(1-t)(1-9t)}$.

n	E_{nn}	n	E_{nn}
0	1	8	9 004 545
1	5	9	76 499 525
2	33	10	653 808 673
3	245	11	5 614 995 765
4	1 921	12	48 416 454 529
5	15 525	13	418 895 174 885
6	127 905	14	3 634 723 102 113
7	1 067 925	15	31 616 937 184 725

Table 4: Numerical values $b_{mk}^{-1/2}$ (alpha=$-1/2$, first 5×5) and $b_{mk}^{+1/2}$ (alpha=$+1/2$, first 5×5), computed to 8 digits. Note: $b_{00}^{-1/2} = \Gamma(1/4)/(3/2)^{1/4}/\sqrt{\pi} \approx 1.8483462$; $b_{00}^{+1/2} = \Gamma(7/4)/(3/2)^{7/4}/\Gamma(3/2) \approx 0.5100825$. Ratio table $F_{mk}^{-1/2} = b_{mk}^{-1/2}/b_{00}^{-1/2}$ has selected exact values $F_{01} = \sqrt{2}/3$, $F_{11} = 4/9$, $F_{22} = 17/54$.

			$\alpha = -1/2$		
$m\backslash k$	0	1	2	3	4
0	1.848346	0.871319	0.586899	0.459225	0.388655
1	0.871319	0.821487	0.592857	0.432962	0.337499
2	0.586899	0.592857	0.581887	0.479110	0.375792
3	0.459225	0.432962	0.479110	0.471595	0.410982
4	0.388655	0.337499	0.375792	0.410982	0.405778
			$\alpha = +1/2$		
$m\backslash k$	0	1	2	3	4
0	0.510082	0.138827	0.010348	−0.022353	−0.025465
1	0.138827	0.302271	0.160526	0.048670	−0.003107
2	0.010348	0.160526	0.223555	0.151909	0.067602
3	−0.022353	0.048670	0.151909	0.184721	0.140266
4	−0.025465	−0.003107	0.067602	0.140266	0.161038